

Paper 2 · Practice Questions — WORKED SOLUTIONS

Mathematics Advanced 12 · Revision — NESAs-standard solutions: state the rule, show every step, give exact answers unless a rounding instruction is given.

Question 1 — integrating $(ax+b)^n$

$$\int \sqrt[3]{(4x+7)} \, dx = \int (4x+7)^{1/3} \, dx$$

Rule: $\int (ax+b)^n \, dx = (ax+b)^{n+1} / [a(n+1)] + C$,
with $a = 4$, $n = 1/3 \Rightarrow n+1 = 4/3$.

$$= \frac{(4x+7)^{4/3}}{4 \times \frac{4}{3}} + C = \frac{(4x+7)^{4/3}}{16/3} + C$$

$$= \frac{3}{16} (4x+7)^{4/3} + C$$

Question 2 — tangent to $y = (\ln x + 1)^3$ at $x = e$

Step 1 — point of contact

$$y(e) = (\ln e + 1)^3 = (1+1)^3 = 8 \Rightarrow (e, 8)$$

Step 2 — gradient (chain rule)

$$\frac{dy}{dx} = 3(\ln x + 1)^2 \times \frac{1}{x} = \frac{3(\ln x + 1)^2}{x}$$

$$m = \frac{3(1+1)^2}{e} = \frac{12}{e}$$

Step 3 — point-gradient form

$$y - 8 = \frac{12}{e}(x - e) = \frac{12}{e}x - 12$$

$$y = \frac{12}{e}x - 4 \text{ (or } 12x - ey - 4e = 0)$$

Question 3 — global extrema on a closed domain

On a closed interval the global maximum/minimum occurs either at a stationary point or at an endpoint — so test both.

$$y' = 3x^2 + 2x - 5 = (3x + 5)(x - 1) = 0$$

$$x = -5/3 \text{ or } x = 1 \text{ — both lie in } [-2, 3]$$

Evaluate at all four candidates

x	-2	-5/3	1	3
y	4	$121/27 \approx 4.48$	-5	19

$$y(-2) = -8 + 4 + 10 - 2 = 4 \text{ (endpoint)}$$

$$y(1) = 1 + 1 - 5 - 2 = -5 \text{ (stationary)}$$

$$y(3) = 27 + 9 - 15 - 2 = 19 \text{ (endpoint)}$$

$$\text{Global max} = 19 \text{ at } x = 3; \text{ Global min} = -5 \text{ at } x = 1$$

Question 4 — reading $y = k \cos(ax)$ from a graph

$$\text{Max } y = 3, \text{ min } y = -3 \Rightarrow \text{amplitude } k = 3$$

Consecutive maxima on the graph are at $x = 0$ and $x = 6\pi$, so one full cycle is 6π .

$$\text{Period } T = \frac{2\pi}{a} = 6\pi \Rightarrow a = \frac{2\pi}{6\pi}$$

$$k = 3, a = 1/3$$

Question 5 — probability without replacement

6 black + 8 white = 14 socks. "Same colour" = BB or WW (mutually exclusive).

$$P(BB) = \frac{6}{14} \times \frac{5}{13} = \frac{30}{182}$$

$$P(WW) = \frac{8}{14} \times \frac{7}{13} = \frac{56}{182}$$

$$P(\text{same}) = \frac{30}{182} + \frac{56}{182} = \frac{86}{182}$$

$$= \frac{43}{91} \approx 0.4725$$

Question 6 — normal distribution, empirical rule

(a) Mean

A 95% interval is symmetric about μ .

$$\mu = \frac{42 + 54}{2}$$

$$(a) \mu = 48 \text{ hours}$$

(b) Standard deviation

68-95-99.7 rule: 95% of data lies within $\mu \pm 2\sigma$.

$$48 + 2\sigma = 54 \Rightarrow 2\sigma = 6$$

$$(b) \sigma = 3 \text{ hours}$$

(c) Percentage between 51 and 57 hours

$$z_1 = \frac{51 - 48}{3} = 1, z_2 = \frac{57 - 48}{3} = 3$$

$$P(0 < Z < 1) = 68\%/2 = 34\%; P(0 < Z < 3) = 99.7\%/2 = 49.85\%$$

$$P(1 < Z < 3) = 49.85\% - 34\%$$

$$(c) \approx 15.85\% \text{ (15.74\% if standard normal tables are used)}$$

Question 7 — proving a trig identity

Work on one side only (LHS) and quote each identity used.

$$\text{LHS} = \operatorname{cosec} \theta - \sin \theta = \frac{1}{\sin \theta} - \sin \theta$$

$$= \frac{1 - \sin^2 \theta}{\sin \theta} \text{ (common denominator)}$$

$$= \frac{\cos^2 \theta}{\sin \theta} \text{ (Pythagorean identity } 1 - \sin^2 \theta = \cos^2 \theta)$$

$$= \cos \theta \times \frac{\cos \theta}{\sin \theta} = \cos \theta \cot \theta = \text{RHS} \blacksquare$$

Question 8 — area between a curve and the x-axis

$$y = x^2(x - 3) = x^3 - 3x^2; \text{ x-intercepts at } x = 0, 3$$

Curve is **below** the axis for $0 \leq x \leq 3$ and **above** for $3 \leq x \leq 4$, so the integrals must be done separately and absolute values taken.

$$\int (x^3 - 3x^2) dx = \frac{x^4}{4} - x^3$$

$$A_1 = \left| \left[\frac{x^4}{4} - x^3 \right]_0^3 \right| = \left| \frac{81}{4} - 27 \right| = \frac{27}{4}$$

$$A_2 = \left[\frac{x^4}{4} - x^3 \right]_3^4 = (64 - 64) - \left(-\frac{27}{4} \right) = \frac{27}{4}$$

$$\text{Total} = \frac{27}{4} + \frac{27}{4} = \frac{54}{4}$$

Area = $\frac{27}{2}$ square units (13.5 u²)

Question 9 — arithmetic series

$$S_n = \frac{n}{2}[2a + (n - 1)d]$$

$$S_6 = 3(2a + 5d) = 78 \Rightarrow 2a + 5d = 26 \dots \textcircled{1}$$

$$S_{10} = 5(2a + 9d) = 210 \Rightarrow 2a + 9d = 42 \dots \textcircled{2}$$

$$\textcircled{2} - \textcircled{1}: 4d = 16 \Rightarrow d = 4$$

$$\text{Sub into } \textcircled{1}: 2a + 20 = 26 \Rightarrow 2a = 6$$

First term $a = 3$, common difference $d = 4$

Question 10 — continuous random variable

(a) Show $k = \frac{2}{3}$

For a pdf the total area is 1.

$$\int_0^{2\pi/3} k \sin(x + \pi/3) dx = 1$$

$$\left[-k \cos(x + \pi/3) \right]_0^{2\pi/3} = -k \cos \pi + k \cos \frac{\pi}{3}$$

$$= k + \frac{k}{2} = \frac{3k}{2} = 1 \Rightarrow k = \frac{2}{3} \blacksquare$$

(b) Cumulative distribution function

$$F(x) = \int_0^x \frac{2}{3} \sin(t + \pi/3) dt = \left[-\frac{2}{3} \cos(t + \pi/3) \right]_0^x$$

$$= -\frac{2}{3} \cos(x + \pi/3) + \frac{2}{3} \cos(\pi/3)$$

(b) $F(x) = 0$ for $x < 0$; $\frac{1}{3} - \frac{2}{3} \cos(x + \pi/3)$ for $0 \leq x \leq 2\pi/3$; 1 for $x > 2\pi/3$

(c) Median (2 d.p.)

The median m satisfies $F(m) = 0.5$.

$$\frac{1}{3} - \frac{2}{3} \cos(m + \pi/3) = \frac{1}{2} \Rightarrow -\frac{2}{3} \cos(m + \pi/3) = \frac{1}{6}$$

$$\cos(m + \pi/3) = -\frac{1}{4} \Rightarrow m + \pi/3 = 1.82348$$

(radians)

$$m = 1.82348 - 1.04720 = 0.77628$$

(c) Median $m \approx 0.78$

Question 11 — bearings: sine rule

(a) Size of $\angle ABC$

The north lines at A and B are parallel.

Bearing of B from A = $076^\circ T \Rightarrow$ co-interior angle at B = $180^\circ - 76^\circ = 104^\circ$

Bearing of C from B = $158^\circ T$ (clockwise from north at B)

Angles at B sum to 360° : $\angle ABC = 360^\circ - (104^\circ + 158^\circ)$

(a) $\angle ABC = 98^\circ$

(b) Distance BC (3 s.f.)

In $\triangle ABC$: $AB = c = 62.6$, $AC = b = 71.6$, $B = 98^\circ$

$$\frac{\sin C}{62.6} = \frac{\sin 98^\circ}{71.6} \Rightarrow \sin C = \frac{62.6 \sin 98^\circ}{71.6} = 0.86579$$

$C = 59.97^\circ$ (the obtuse option 120.03° is rejected — it would make A negative)

$$A = 180^\circ - (98^\circ + 59.97^\circ) = 22.03^\circ$$

$$BC = \frac{71.6 \sin 22.03^\circ}{\sin 98^\circ} = 27.117 \text{ km}$$

(b) $BC \approx 27.1 \text{ km}$

Question 12 — future value of an annuity (table)

$$\text{Monthly rate } r = \frac{0.042}{12} = 0.00350$$

$$\text{Number of periods } n = 8 \times 12 = 96$$

From the table, row $n = 96$, column 0.00350: future value factor = 113.86239.

$$FV = M \times \text{factor} \Rightarrow 40\,000 = M \times 113.86239$$

$$M = \frac{40\,000}{113.86239} = 351.3012\dots$$

Heidi must deposit \$351.30 each month

Check: $351.30 \times 113.86239 = \$39\,999.86 \checkmark$ (by contrast \$351.21 gives only \$39 989.61, about \$10 short of the target).